Proposition 10: a factor 2 gap for the cut distance

Proof overview of the Lean 4/Mathlib formalisation

Folder `GsimOT/Proposition12_CutLowerBound`

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport.*
File `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository; statement, equation, page and line numbers refer to this PDF. `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

1 Statement

Paper (Section 3.3, p. 9, l. 324; Figure 3, p. 10; proof in App. E.1, pp. 30–32, l. 950–1001). *There are graphs* G, H *on* 145 *vertices with* $d_{\square}(G,H) = 8$ *and* $d_{\square}^{\mathrm{OT}}(G,H) \leq 4$.

Lean.

- `GsimOT.Prop12.proposition12`: $|V| = 145$, $d_{\square}(G,H) = 8$ and $d_{\square}^{\mathrm{OT}}(G,H) \leq 4$ for the concrete graphs `G12`, `H12` on the vertex type `Vtx`. This is *exactly* the statement of Proposition 10. There are no hypotheses.
- `GsimOT.Prop12.dOTCut_eq_four`: $d_{\square}^{\mathrm{OT}}(G,H) = 4$. This is the Lean certificate of the remark after Proposition 10 (p. 10, l. 330); `GsimOT.Prop12.four_le_costCut` shows $C_{\square}(Q) \geq 4$ for every $Q \in \mathrm{DS}(V,W)$.
- `GsimOT.Prop12.theorem4_item3`: $\delta_{\square}(G,H) = 2\,\delta_{\square}^{\mathrm{OT}}(G,H)$, Theorem 3 (3) (p. 4, l. 126).

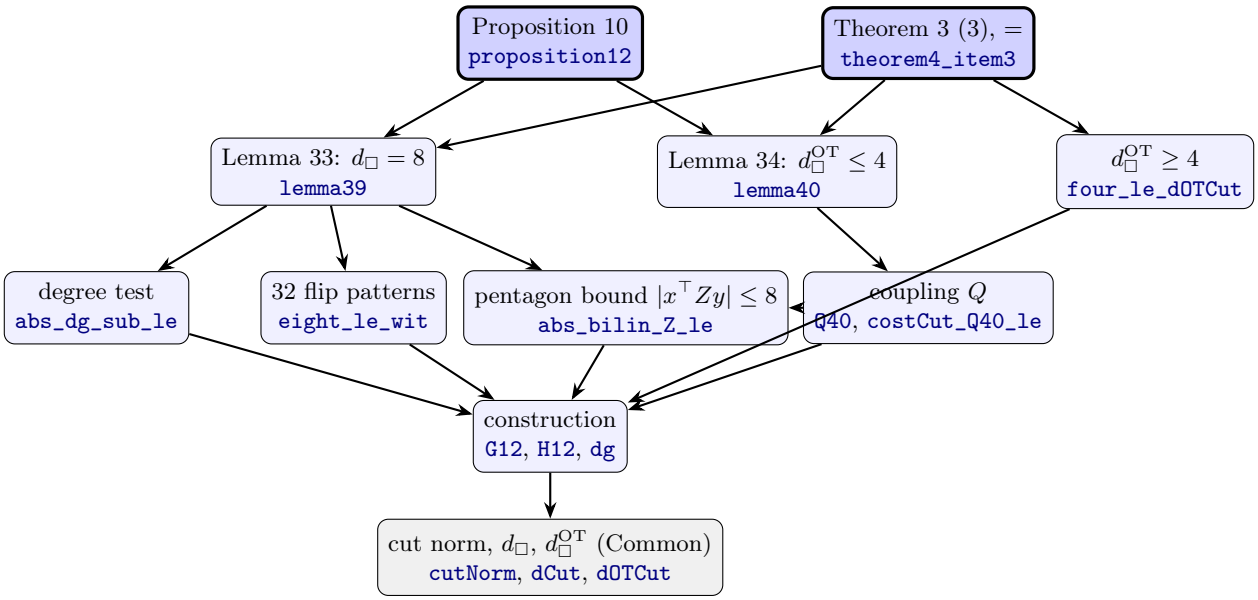
Here $d_{\square}(G,H) = \min_{\pi} \|A_G^{\pi} - A_H\|_{\square}$ (`dCut`), $d_{\square}^{\mathrm{OT}}(G,H) = \inf_{Q \in \mathrm{DS}(V,W)} C_{\square}(Q)$ (`dOTCut`) with $C_{\square}(Q) = \|B(G,H,Q)\|_{\square}$ (`costCut`), and $\delta = d/n^2$.

The equality claim. The remark after Proposition 10 (p. 10, l. 330) says: "Note that one of our Lean certificates verifies that the $d_{\square}^{\mathrm{OT}}$-bound is tight, so $d_{\square}^{\mathrm{OT}}(G,H) = 4$." Its footnote 2 (p. 10) links the certificate, the file `GsimOT/Proposition12_CutLowerBound/OptimalAlignment.lean` of this folder. The paper refers to this file for the proof; the argument is outlined below.

2 Where the proof lives

File	Content
`Construction.lean`	vertex type, pentagon matrix Z, graphs `G12`, `H12`, adjacency entries, degrees, the finite pentagon bound $\lvert x^{\top} Z y\rvert \leq 8$ (App. E.1, pp. 30–31)
`Lemma39_BijectionCost.lean`	Lemma 33: $d_{\square}(G,H) = 8$ (`lemma39`)
`Lemma40_FractionalCost.lean`	Lemma 34: $d_{\square}^{\mathrm{OT}}(G,H) \leq 4$ (`lemma40`)
`OptimalAlignment.lean`	$C_{\square}(Q) \geq 4$ for all Q (`four_le_costCut`), $d_{\square}^{\mathrm{OT}}(G,H) = 4$ (`dOTCut_eq_four`); the Lean certificate of the remark after Proposition 10
`Proposition12.lean`	Proposition 10 (`proposition12`) and Theorem 3 (3) (`theorem4_item3`)

3 Proof layout



4 The proof step by step

Construction (App. E.1, pp. 30–31, l. 950–966). The core consists of five pairs $\{v_i^+, v_i^-\}$, $i \in \mathbb{Z}_5$, and each pair i has its own set L_i of $9(i+1)$ markers adjacent to both vertices of the pair; $135 + 10 = 145$ vertices (`card_Vtx`). Core edges follow Z ($Z_{ij} = +1$ on the sides, -1 on the diagonals of the pentagon): $v_i^\sigma v_j^\tau \in E(G)$ iff $\sigma\tau = Z_{ij}$, $w_i^\sigma w_j^\tau \in E(H)$ iff $\sigma\tau = -Z_{ij}$ (`gAdj`, `hAdj`, `G12`, `H12`). Both graphs live on one vertex type `Vtx`; the natural bijection id ($v_i^\sigma \mapsto w_i^\sigma$, $L_i \to L_i'$; p. 31, l. 984–985) is `Equiv.refl`. The degrees are $13, 22, 31, 40, 49$ on the pairs and 2 on markers, in both graphs (`sum_adj_G12`, `sum_adj_H12`); distinct degrees differ by at least 9 (`nine_le_of_dg_ne`).

Lemma 33, $d_\square(G, H) = 8$ (p. 31, l. 967; proof l. 968–988).

1. *Lower bound, markers fix the pairs* (l. 968–972). With $S = \{\pi(v)\}$, $T = W$ one gets $C_\square(\pi) \ge |\deg_G(v) - \deg_H(\pi v)|$ (`abs_dg_sub_le`), which is ≥ 9 unless π preserves all degrees. A degree-preserving π maps every core vertex into its own pair (`symm_inl_of_dg`), i.e. $\pi(v_i^\sigma) = w_i^{r_i\sigma}$.
2. *Lower bound, flip patterns* (l. 973–984). On the core, $A_G^\pi - A_H$ has entries $M_{ij}\sigma\tau$ with $M_{ij} = \frac{1}{2}Z_{ij}(1 + r_i r_j)$ (`Mr`), and core cuts realise $x^\top M y$ for all $x, y \in \{-1, 0, 1\}^5$ (`core_sum_eq_bilin`, `cutOf`). The paper checks $\|M\|_{\infty\to 1} \ge 8$ for the cases $|K| \le 2$ up to symmetry; Lean checks all 32 patterns r with an explicit table of maximisers `wit` by `decide` (`eight_le_wit`, `eight_le_of_dg`). Together: `eight_le_cutNorm`.
3. *Upper bound, π = id* (l. 984–988). $A_G - A_H$ vanishes outside the core, every cut value is $x^\top Z y$ (`cut_sum_eq_core`), and $|x^\top Z y| \le 8$ (`abs_bilin_Z_le`, reduced to $\sum_j |(Zx)_j| \le 8$ over the 3^5 vectors x, `sum_abs_Z_mul_le_int`). Hence $C_\square(\mathrm{id}) \le 8$ (`cutNorm_refl_le`).

Lemma 34, $d_\square^{\mathrm{OT}}(G, H) \le 4$ (p. 31, l. 989; proof pp. 31–32, l. 990–1000). Q is the identity on markers and $\frac{1}{2}$ on each pair block (`Q40`, `Q40_isDS`). Discrepancies with a marker atom vanish (`B40_eq_zero_left`, `B40_eq_zero_right`); on the 20 core atoms the entry of $B(G, H, Q)$ is $\frac{1}{2} \cdot \frac{1}{2}Z_{ij}(\sigma\sigma' + \tau\tau') \cdot \frac{1}{2}$ (`B40_atom`). Splitting into straight and crossed atoms (`atom_sum_eq`) every cut value is a sum of two terms $\frac{1}{4}x^\top Z y$, so $C_\square(Q) \le \frac{1}{4} \cdot 8 + \frac{1}{4} \cdot 8 = 4$ (`costCut_Q40_le`), and $d_\square^{\mathrm{OT}} \le C_\square(Q)$ (`lemma40`).

$d_\square^{\mathrm{OT}}(G, H) \ge 4$ (remark after Proposition 10, p. 10, l. 330; `OptimalAlignment.lean`). Fix $Q \in \mathrm{DS}(V, W)$, write $q_x = Q(v, w)$ for atoms $x = (v, w)$, $d(x) = \deg_G(v) - \deg_H(w)$, $L = \sum_x q_x|d(x)|$ and $B(f, g) = \sum_{x,y} f(x)q_x D(x, y)q_y g(y)$.

1. *Fractional cuts*: $|B(f, g)| \le C_\square(Q)$ for $f, g \colon \Omega \to [0, 1]$. A linear form on $[0, 1]^\Omega$ is extremal at an indicator (`exists_abs_sum_mul_le`); apply this in f, then in g (`abs_bilin_le_cutNorm`).
2. *Degree identities*: $\sum_y D(x, y)q_y = d(x)$ (`sum_disc_mul`, `sum_B_right`) and $\sum_x q_x d(x) = 0$ (`sum_mul_dd`).

3. *Core test and degree discrepancy* (`degree_add_bil_le`): for $|s|, |t| \le 1$ with $s = 0$ where $d \neq 0$, and $e = \operatorname{sgn} d$, the pairs $f_\pm = \frac{1}{2}(1 + e \pm s)$, $g_\pm = \frac{1}{2}(1 \pm t)$ take values in $[0,1]$ and $B(f_+, g_+) + B(f_-, g_-) = \frac{1}{2}(L + B(s,t))$, so $L + B(s,t) \le 4C_\square(Q)$.

4. *Mass leaving the pairs*: mass sent from a core vertex outside its pair has $|d| \ge 9$, so $L \ge 9(10 - T)$, where T is the mass on the 20 core atoms (`nine_mul_le_row`, `nine_mul_le_L`).

5. *The pentagon* (`exists_zForm_ge`): for $x \in [0,2]^5$ there are $a, b \in [-1,1]^5$ with $a^\top \operatorname{diag}(x) Z \operatorname{diag}(x) b \ge \frac{16}{5} \sum_i x_i - 8$: for $x_k = \max x$ and $a_j = Z_{kj}$ ($j \neq k$), flipping a_k changes the quadratic form by $4x_k(S - x_k)$ (`zForm_flip`, by `fin_cases` on k), and $2M(S - M) \ge \frac{16}{5}S - 8$ for $M \le S \le 5M$, $M \le 2$.

6. *Test functions*: $s = \sigma a_i$ on straight atoms (v_i^σ, w_i^σ), $\sigma a'_i$ on crossed atoms (`testFn`); then $B(s,t) = a^\top C Z C b + a'^\top H Z H b'$ with c_i, h_i the straight and crossed masses of pair i (`disc_atom`, `bil_testFn`). Item 5 for c and h gives $B(s,t) \ge \frac{16}{5}T - 16$.

7. Together $4C_\square(Q) \ge 9(10 - T) + \frac{16}{5}T - 16 \ge 16$ since $T \le 10$ (`four_le_costCut`), hence $d_\square^{\mathrm{OT}} \ge 4$ (`four_le_dOTCut`) and, with Lemma 34, $= 4$ (`dOTCut_eq_four`).

Proposition 10 and Theorem 3 (3). `proposition12` combines the two lemmas, as the proof of Proposition 10 does (p. 32, l. 1001). `theorem4_item3` divides by 145^2: $\delta_\square = 8/145^2 = 2 \cdot 4/145^2 = 2\,\delta_\square^{\mathrm{OT}}$.

5 Assumptions and trusted base

No hypotheses. The finite checks use `decide` (no `native_decide`). Only the standard axioms `propext`, `Classical.choice`, `Quot.sound` are used.

6 Notes on the formalisation

- The case analysis on $|K|$ in Lemma 33 is replaced by a check of all 32 flip patterns with explicit witnesses.
- $d_\square^{\mathrm{OT}} \ge 4$ is proved in `OptimalAlignment.lean`, the Lean certificate to which the remark after Proposition 10 refers (p. 10, l. 330 and footnote 2); its proof is outlined above.
- Both graphs live on one vertex type `Vtx`, so the natural bijection id is `Equiv.refl`.

7 How to check this result

```
lake build GsimOT.Proposition12_CutLowerBound.Proposition12
lake env lean scripts/CheckAxioms.lean      # prints the axioms of proposition12, dOTCut_eq_four
```

The correspondence table is `Proposition12_correspondence.pdf` in the same folder.